

$$\text{RIP} \Rightarrow \text{NSP} \quad A$$

RIP_(s):

$$\underline{X}^T \underline{\Theta} = \underline{I}$$

$$\left\| \underbrace{X_S^T X_S}_n - I \right\|_2 < 1/3$$

$$S \subseteq \{1, \dots, d\}$$

$$|S| \leq s$$

Spectral Norm $\longrightarrow$ $\|A\|_2 := \sigma_{\max}(A)$

max Singular Val for a matrix A

$$\sigma_{\max}(A) :=$$

$$\max_{v \in S^{n-1}} \|Av\|_2$$

min Singular Val

$$\sigma_{\min}(A) :=$$

$$\min_{v \in S^{n-1}} \|Av\|_2$$

$$\begin{aligned} & \xrightarrow{\frac{v}{\|v\|_2}} \equiv \max_{v \in \mathbb{R}^n} \left\| A \frac{v}{\|v\|_2} \right\|_2 \\ & = \max_{v \in \mathbb{R}^n} \frac{\|Av\|_2}{\|v\|_2} \\ & \quad \|Av\|_2 = \sqrt{v^T A^T A v} \end{aligned}$$

$$\|A\|_2 = \max_{v \in \mathbb{S}^{n-1}} |v^T A v| \quad \text{if } A_{n \times n}$$

$$\gamma_{\max}(A) := \max_{v \in \mathbb{S}^{n-1}} v^T A v$$

$$\|A\|_2 = \max \{ \gamma_{\max}(A), \underbrace{|\gamma_{\min}(A)|} \}$$

$$\gamma_{\min}(A) := \min_{v \in \mathbb{S}^{n-1}} v^T A v$$

$$A = U^T \Sigma V$$

? What is the relationship of $\sigma_{\max}$, $\sigma_{\min}$, $\gamma_{\max}$, $\gamma_{\min}$ for a square matrix?

SVD

$$\begin{aligned} \downarrow A &= A^T \\ \left\{ \begin{aligned} v^T A^T &= \lambda v^T \\ A v &= \lambda v \\ A^T A v &= \lambda' v \end{aligned} \right. \end{aligned}$$

$$\lambda' = \lambda^2$$

$$\sigma_{\max}(A) = \max_{v \in \mathbb{S}^{n-1}} \sqrt{v^T \underbrace{A^T A}_B v}$$

$$= \sqrt{\gamma_{\max}(B)} \gamma_{\max}(A) \\ \stackrel{?}{=} \gamma_{\max}(A)$$

$$\sigma_{\max}(A) = \max_{\substack{V \in \mathbb{R}^{n-1} \\ W \in \mathbb{R}^{n-1}}} \sqrt{V^T V^T \underbrace{\Sigma U U^T \Sigma}_{\substack{I \\ V^T V^T \Sigma^2 V}} V} = \left\{ \begin{array}{l} \max \\ \sigma_{\min}(A) \\ \sigma_{\max}(A) \end{array} \right\}$$

$$\Sigma = \begin{bmatrix} \sigma_{\max} & & \\ & \ddots & \\ & & \sigma_{\min} & \ddots \end{bmatrix}$$

$$A^{\text{square}} = \underbrace{U^T}_{\substack{= \\ W^T}} \underbrace{\Sigma}_{=} \underbrace{V}_{\substack{= \\ W}}$$

$$A V = V \Lambda \Rightarrow V^T A V = \Lambda \rightarrow A = V \Lambda V^T$$

Th. If X satisfies RIP Cond of order $2s$ of Const. $< 1/3$
 then $X_{\wedge}$ satisfy RNSP. $|S| \leq s$.
 would

Proof: $\mathbf{0} \neq \boldsymbol{\theta} \in \mathcal{N}(X)$ RIP $\Rightarrow \underbrace{\|\boldsymbol{\theta}_S\|_1}_{\text{زفر}} < \underbrace{\|\boldsymbol{\theta}_{S^c}\|_1}_{\text{حکم}}$.

$$\tilde{\boldsymbol{\theta}}_S = \begin{bmatrix} \theta_1 \\ 0 \\ \theta_2 \\ 0 \\ \vdots \end{bmatrix} := \begin{cases} \theta_j & j \in S \\ 0 & \text{otherwise} \end{cases} \quad \|\tilde{\boldsymbol{\theta}}_S\|_1 = \|\boldsymbol{\theta}_S\|_1$$

$$\{1, \dots, d\} \overset{S}{=} \underbrace{(S_0)}_{\substack{\uparrow \\ \text{contain } S \\ \text{max abs. elements} \\ \text{of } \boldsymbol{\theta}}} \cup \underbrace{S_1}_{\uparrow} \cup S_2 \cup \dots \cup \underbrace{S_k}_{\uparrow} \quad \boldsymbol{\theta} = \tilde{\boldsymbol{\theta}}_{S_0} + \tilde{\boldsymbol{\theta}}_{S_1} + \dots + \tilde{\boldsymbol{\theta}}_{S_k}$$

$$\boldsymbol{\theta} \in \mathcal{N} \Rightarrow X\boldsymbol{\theta} = \mathbf{0}$$

$$\Rightarrow X(\tilde{\boldsymbol{\theta}}_{S_0} + \dots + \tilde{\boldsymbol{\theta}}_{S_k}) = \mathbf{0} \Rightarrow$$

$$\underbrace{X\tilde{\boldsymbol{\theta}}_{S_0}} = - \underbrace{X\left(\sum_{i=1}^k \tilde{\boldsymbol{\theta}}_{S_i}\right)}$$

$$\text{RIP} \Rightarrow \|\tilde{\theta}_{S_0}\|_2^2 \leq \frac{1}{1-\delta_{2S}} \left\| \frac{1}{\sqrt{n}} X^T \tilde{\theta}_{S_0} \right\|_2^2$$

$$S_0 = S \quad \underbrace{\left\| \frac{X_S^T X_S}{n} - I \right\|_2}_A \leq \underline{\delta_{2S}} \quad \rightarrow \tilde{\theta}_S^T \underbrace{\frac{X^T X}{n}} \tilde{\theta}_S$$

$$\max_{v \in \mathbb{S}^{n-1}} |v^T A v| = \max_{v \in \mathbb{S}^{n-1}} \left| v^T \left(\frac{X_S^T X_S}{n} - I \right) v \right|$$

$$= \max_{v \in \mathbb{S}^{n-1}} \left| \tilde{\theta}_S^T \frac{X_S^T X_S}{n} v - \|v\|_2^2 \right|$$

$$\begin{aligned} -\left| \tilde{\theta}_S^T \frac{X_S^T X_S}{n} \tilde{\theta}_S \right| + \|\tilde{\theta}_S\|_2^2 &\leq \left| \tilde{\theta}_S^T \frac{X_S^T X_S}{n} \tilde{\theta}_S - \|\tilde{\theta}_S\|_2^2 \right| \leq \delta_{2S} \|\tilde{\theta}_S\|_2^2 \\ \Rightarrow \underbrace{\left| \tilde{\theta}_S^T \frac{X_S^T X_S}{n} \tilde{\theta}_S \right|}_{\geq} &\geq (1 - \delta_{2S}) \|\tilde{\theta}_S\|_2^2 \end{aligned}$$

$$\Rightarrow \|\tilde{\Theta}_{s_0}\|_2^2 \leq \frac{1}{1 - \delta_{2s}} \underbrace{\left\| \frac{1}{\sqrt{n}} X \tilde{\Theta}_{s_0} \right\|_2^2}_{\tilde{\Theta}_{s_0}^T \underbrace{X^T X}_n \tilde{\Theta}_{s_0}}$$

$$\sum_{i=1}^k - X \tilde{\Theta}_{s_i}$$

$$= \left| \frac{1}{1 - \delta_{2s}} \sum_{i=1}^k \frac{1}{n} \langle X \tilde{\Theta}_{s_0}, X \tilde{\Theta}_{s_i} \rangle \right|$$

$$= \frac{1}{1 - \delta_{2s}} \left| \sum_{i=1}^k \underbrace{\tilde{\Theta}_{s_0}^T}_V \underbrace{\left[\frac{X^T X}{n} - I \right]}_W \underbrace{\tilde{\Theta}_{s_i}}_W \right| \leq \frac{1}{1 - \delta_{2s}} \sum_{i=1}^k \underbrace{\|\tilde{\Theta}_{s_0}\|_2}_{\leq \delta_{2s}} \cdot \underbrace{\|A \tilde{\Theta}_{s_i}\|_2}_{\leq \underbrace{\sigma_{\max}(A)}_{\leq \delta_{2s}} \|\tilde{\Theta}_{s_i}\|_2}$$

$$\Rightarrow \frac{1}{\sqrt{s}} \|\tilde{\Theta}_s\|_1 \leq \|\tilde{\Theta}_s\|_2 \leq \frac{\delta_{2s}}{1-\delta_{2s}} \sum_{i=1}^k \|\tilde{\Theta}_{s_i}\|_2 \leq \frac{\delta_{2s}}{1-\delta_{2s}} \sum_{i=1}^k \frac{1}{\sqrt{s}} \|\tilde{\Theta}_{s_i}\|_1$$

$$\|\tilde{\Theta}_{s_i}\|_\infty \leq \frac{1}{s} \|\tilde{\Theta}_{s_{i-1}}\|_1$$

$$\begin{aligned} \|\tilde{\Theta}_{s_i}\|_2 &\leq \sqrt{s} \|\tilde{\Theta}_{s_i}\|_\infty \\ &\leq \frac{\sqrt{s}}{s} \|\tilde{\Theta}_{s_{i-1}}\|_1 \\ &\quad \downarrow \\ &\quad \frac{1}{\sqrt{s}} \end{aligned}$$

$$\leq \frac{\delta_{2s}/\sqrt{s}}{1-\delta_{2s}} \left[\|\tilde{\Theta}_{s_0}\|_1 + \sum_{j=1}^k \|\tilde{\Theta}_{s_j}\|_1 \right]$$

$$\delta_{2s} < 1/3$$

$$\Rightarrow \|\tilde{\Theta}_s\|_1 \leq \frac{\delta_{2s}}{1-\delta_{2s}} [\|\tilde{\Theta}_s\|_1 + \|\tilde{\Theta}_{s^c}\|_1] \quad \tilde{\Theta}_{s^c}$$

$$\frac{\delta_{2s}}{1-\delta_{2s}} > 1/2 \Rightarrow \|\tilde{\Theta}_s\|_1 < \|\tilde{\Theta}_{s^c}\|_1 \quad \square$$

$$X_{ij} \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

$$n = O(\underbrace{S \log d/s}) \Rightarrow \text{with high prob.}$$

$$X \in \text{RIP}(2s, 1/3)$$